

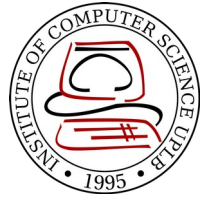
CMSC 137

Data Communications and Networking

Lecture Module
18 – Network Layer



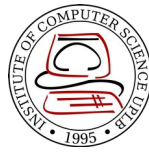
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LOS BAÑOS



Module 18: Network Layer

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Lecturer

Introduction

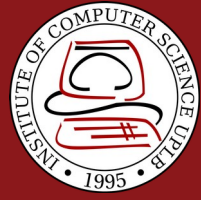


- The network layer is responsible for the delivery of the individual packets from the source to the destination host
- Duties:
 - **Packetizing** – encapsulating the payload in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination
 - **Routing** – finding the best one among possible routes
 - **Forwarding** – action applied by each router when a packet arrives at one of its interfaces

Logical Addressing



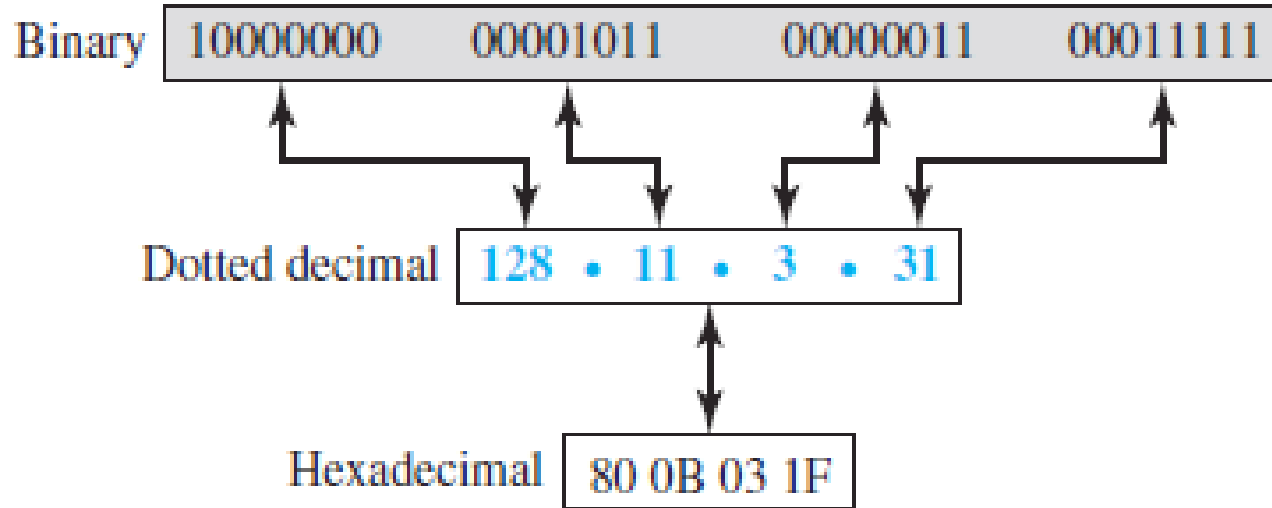
A universal addressing system in which each host can be identified uniquely regardless of the underlying physical network



IPv4 Address

- RFC 791
- 32-bit addresses that *uniquely* and *universally* defines the connection of a host or a router
- Address of the connection, not the host or router, because if the device is moved to another network, the IP address may be changed
- has 2^{32} or 4,294,967,296 address space
- partitioned into four groups of eight bits (called octets)
- each octet is treated as independent unit

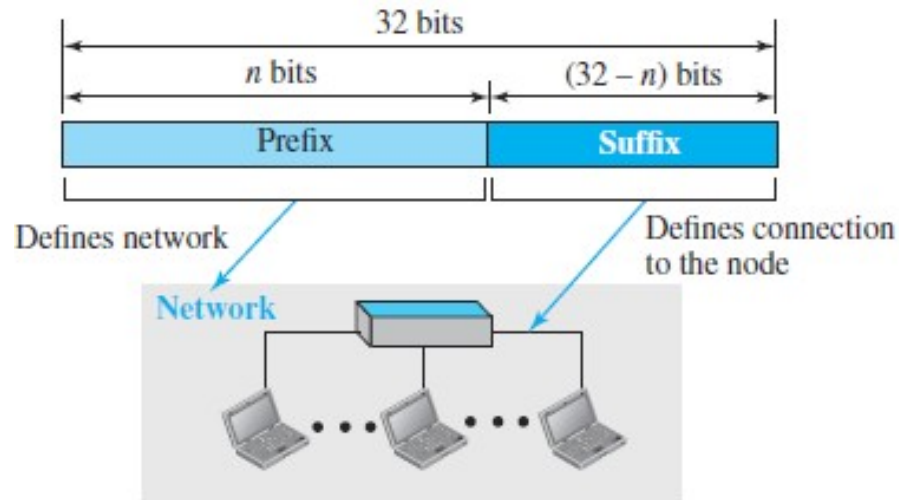
IPv4 Address Notation



Hierarchy in Addressing



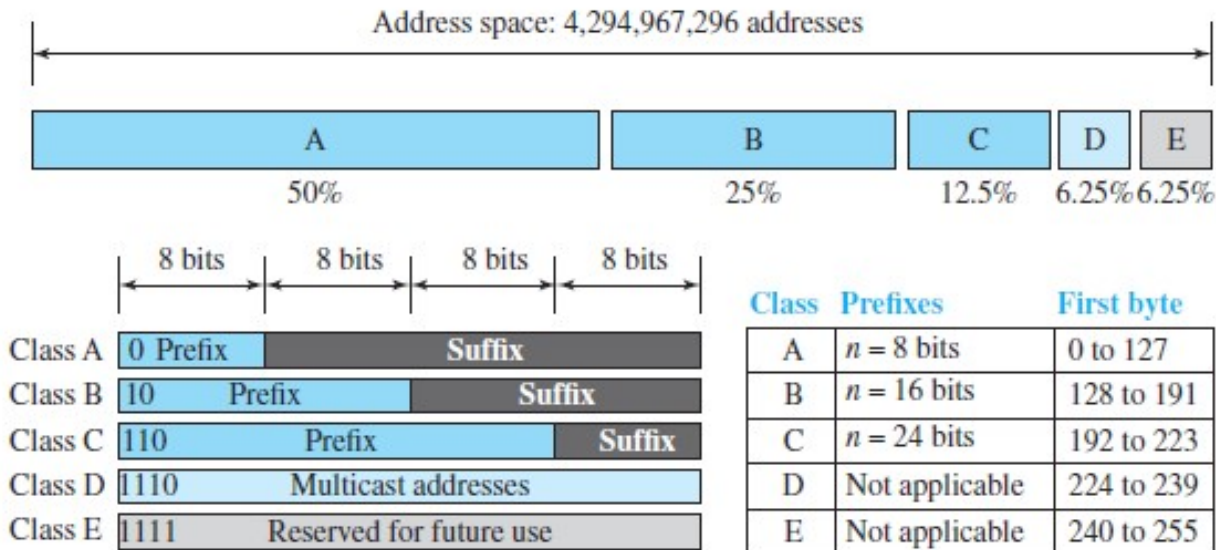
- A 32-bit IPv4 address is hierarchical divided into 2 parts:
 - Prefix – defines the network
 - Suffix – defines the node



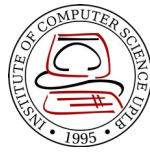
Classful Addressing



- Address space is divided into 5 classes: A, B, C, D and E
- Classification is determined by the value of the first four bits (bits 0 through 3)



Classful Addressing



IP Class	A	B	C	D	E
Format	N.H.H.H	N.N.H.H	N.N.N.H	N/A	N/A
High order bits	0	10	110	1110	11110
Address Range	1.x.x.x to 126.x.x.x	128.x.x.x to 191.x.x.x	192.x.x.x to 223.x.x.x	224.x.x.x to 239.x.x.x	240.x.x.x to 255.x.x.x
No of bits for network/host	7/24	14/16	21/8	Not for commercial use	N/A
Number of network (block)	128	16,384	2,097,152	1	1
Number of host	$2^{24} - 2 = 16777214$	$2^{16} - 2 = 65534$	$2^8 - 2 = 254$	(268,435,456)	(268,435,456)
Netmask	255.0.0.0	255.255.0.0	255.255.255.0		
Purpose	Few large organizations	Medium size organizations	Relatively small organizations	Multicast groups	Experimental

Address Depletion



- Since addresses were not distributed properly, the Internet was faced with the problem of the addresses being rapidly used up, resulting in no more addresses available for organizations and individuals that needed to be connected to the Internet
 - Class A addresses can be only assigned to 128 organizations in the world with 16,777,216 nodes
 - Class E addresses were almost never used

Subnetting/Supernetting



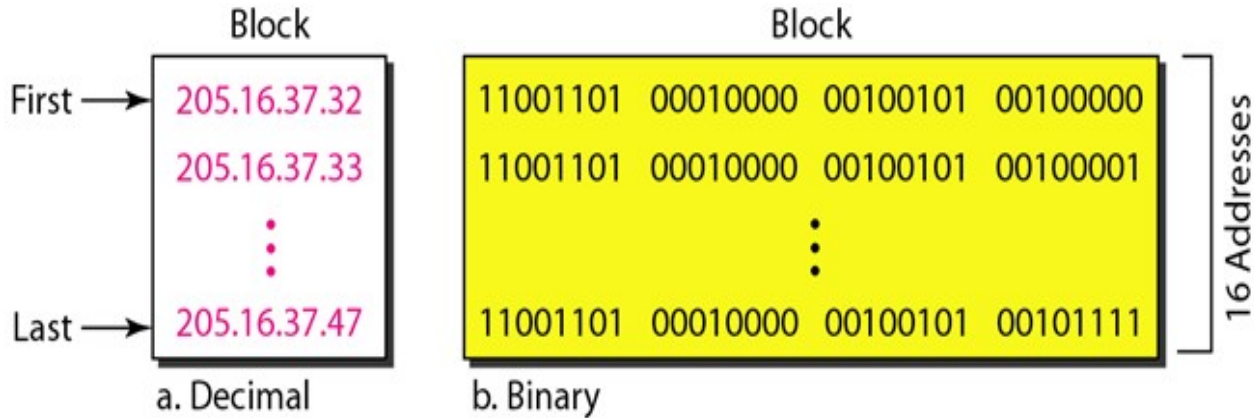
- Subnetting
 - Divide addresses (Class A and B) into several contiguous groups and assign each group to smaller groups (subnet) or share part of address with neighbors
 - Did not work since most large organizations were not happy dividing the block
- Supernetting
 - Combine several class C blocks to create larger range of addresses
 - Did not work since it makes the routing of packets more difficult

Classless Addressing



- To overcome address depletion and give more organization access to the Internet, the class privilege was removed
- Whole address space is divided into variable length blocks
- The prefix defines the block (network); the suffix defines the node
- The size of network is inversely proportional to the length of the prefix
- **Restrictions:**
 - The addresses in a block must be contiguous, one after another
 - The number of addresses in a block must be a power of 2
 - The first address must be evenly divisible by the number of addresses

Classless Addressing



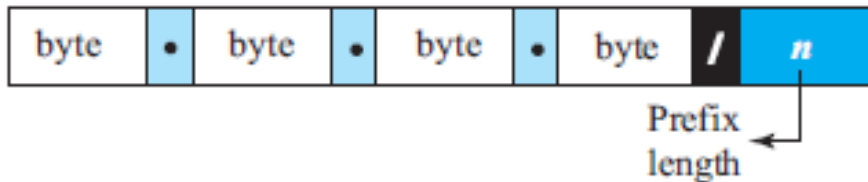
- Address is contiguous
- Number of address is a power of 2 ($16 = 2^4$)
- First address is divisible by 16
- First address when converted to a decimal number is 3,440,387,360

Slash Notation



Question: How to find the prefix length if an address is given?

- Since the prefix length is not inherent in the address, it is needed to give the prefix length separately
- The prefix length n is added to the address separated by a slash
- Formally referred as the [classless interdomain routing](#) or CIDR (informally slash notation)



Examples:

12.24.76.8/8

23.14.67.92/12

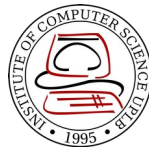
220.8.24.255/25

Extracting Information from Address



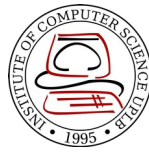
- Given any address in the block, we would like to know the 3 pieces of info: the first address in the block, the last address and number of addresses
 - The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s
 - The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s
 - The number of addresses in the block can be found using the formula 2^{32-n}

Extracting Information from Address



- **Question:** A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28
 - What is the first address in the block?
 - What is the last address in the block?
 - Find the number of addresses in the block?

Extracting Information from Address



- Binary representation of 205.16.37.39

11001101 00010000 00100101 00100111

- Set 32 – 28 rightmost bits to 0

11001101 00010000 00100101 00100000

or 205.16.37.32

- Set 32 – 28 rightmost bits to 1

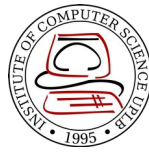
11001101 00010000 00100101 00101111

or 205.16.37.47

- Value of n is 28, so 2^{32-28}

$$2^4 = 16$$

Address Mask



- Another way to find the first and last address
- 32-bit number in which the n leftmost bits are set to 1s and the rest of the bits ($32 - n$) are set to 0s
- A computer can easily find the address mask because it is the complement of $(2^{32-n} - 1)$
- A computer program can extract information using the 3 bit-wise operations NOT, AND and OR
 - Number of address in the block $N = \text{NOT}(\text{mask}) + 1$
 - 1st address = $(\text{any address in the block}) \text{ AND } (\text{mask})$
 - Last address = $(\text{any address}) \text{ OR } [\text{NOT}(\text{mask})]$

Address Mask



- Previous sample: 205.16.37.39/28
- Binary representation is

11001101 00010000 00100101 00100111

- Address Mask: 11111111 11111111 11111111 11110000

- Number of address:

NOT(11111111 11111111 11111111 11110000) + 1

00000000 00000000 00000000 00001111 + 1

15 + 1 = 16

- First address

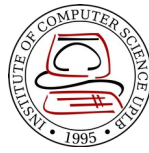
11001101 00010000 00100101 00100111

AND 11111111 11111111 11111111 11110000

11001101 00010000 00100101 00100000

205.16.37.32

Address Mask



- Previous sample: 205.16.37.39/28
- Binary representation is

11001101 00010000 00100101 00100111

- Address Mask: 11111111 11111111 11111111 11110000

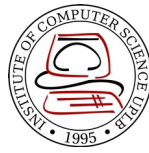
- Last Address

11001101 00010000 00100101 00100111
OR 00000000 00000000 00000000 00001111

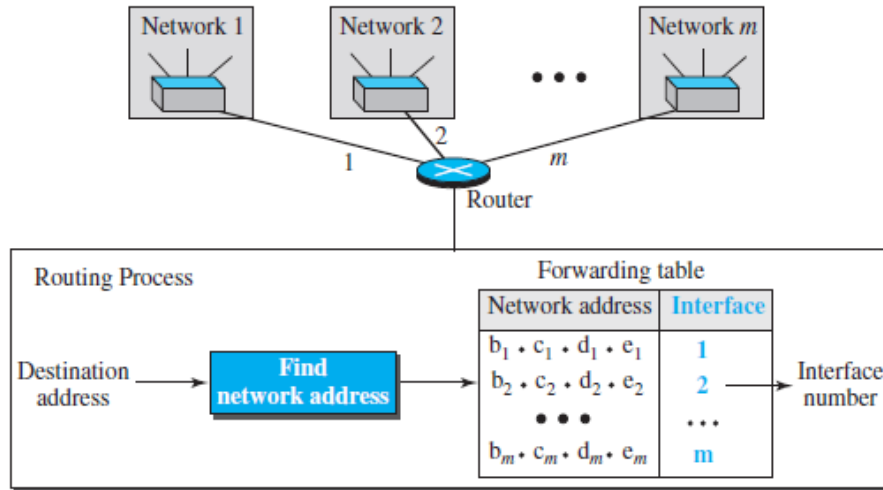
00111101 00010000 00100101 00101111

205.16.37.47

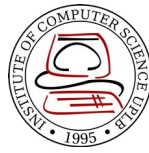
Network Address



- When organization is given a block of address, it is free to allocate the addresses to devices that need to be connected to the Internet
- The first address (normally) is treated as a special address; it is used in routing a packet to its destination network and defines the organization to the rest of the world

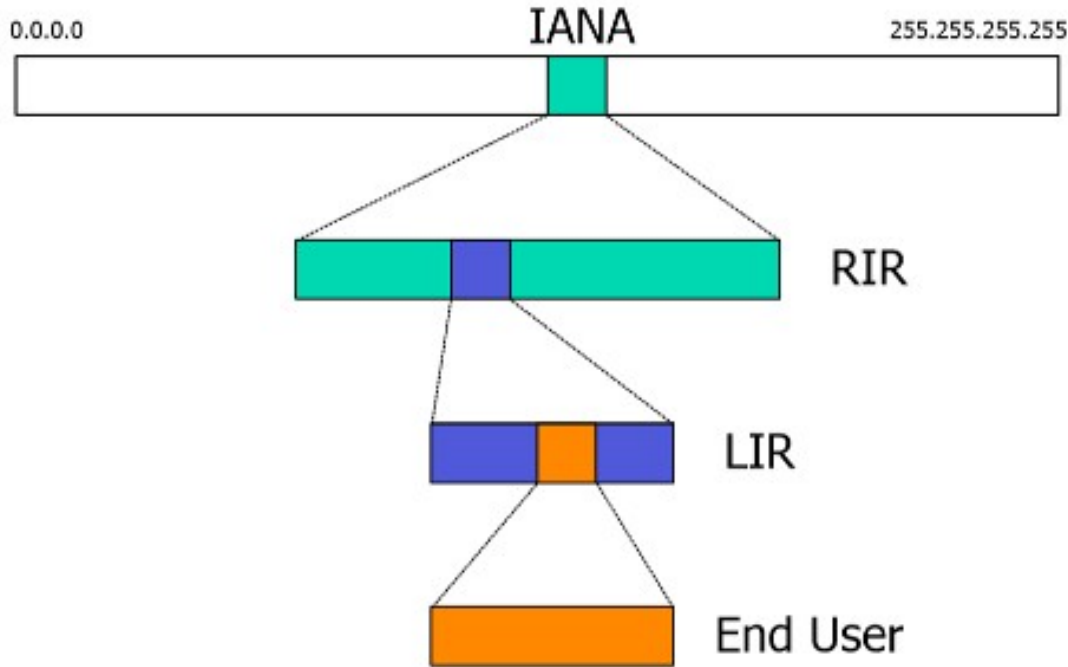


Address Allocation



- [Internet Corporation for Assigned Names and Numbers \(ICANN\)](#) – global authority for address allocation
- [Internet Assigned Numbers Authority \(IANA\)](#)
 - Manages the DNS root, the .int and .arpa domains
 - Coordinates the global pool of IP and AS numbers, providing them to Regional Internet Registries
 - Manages internet protocol's numbering system

Address Allocation



Regional Internet Registries

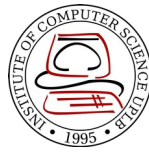
- AFRINIC
- ARIN
- APNIC
- LACNIC
- RIPE NCC

Special Addresses

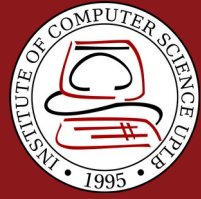


- **This-host address**
 - Only address in the block $0.0.0.0/32$. It is used whenever a host needs to send an IP datagram but it does not know its own address to use as the source address
- **Limited-broadcast address**
 - Only address in the block $255.255.255.255/32$. It is used whenever a router needs to send a datagram to all devices in a network
- **Loopback address**
 - The block $127.0.0.0/8$. A packet with one of the addresses in this block as the destination address will never leave the host

Special Addresses



- Private addresses
 - 4 blocks are assigned as private addresses: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16 and 169.254.0.0/16
 - Used in NAT
- Multicast address
 - The block 224.0.0.0/4

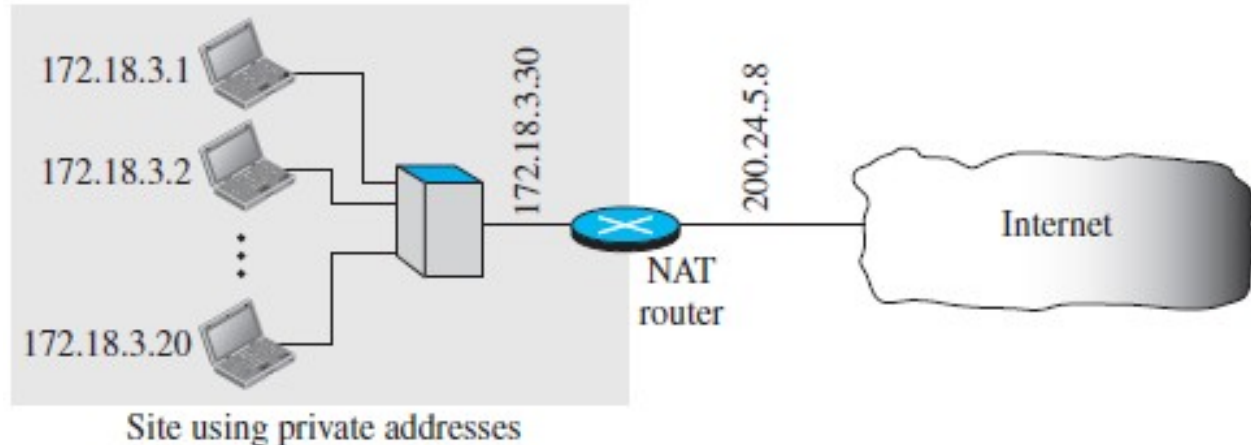


Network Address Translation

Network Address Translation (NAT)



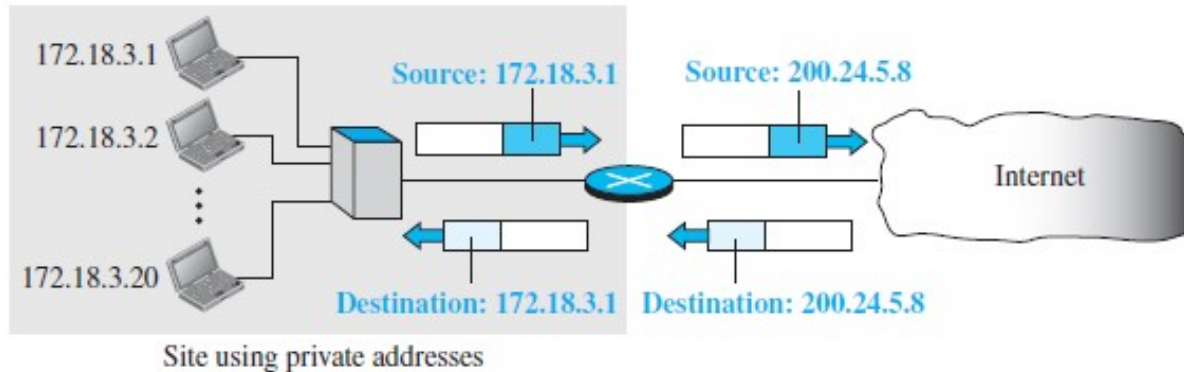
- Technology that can provide the mapping between the private and universal address, and at the same time support virtual private networks
- Allows a site to use a set of private addresses for internal communication and a set of global Internet addresses (at least one) for communication with the rest of the world
- **RFC 3022**



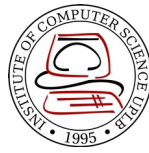
NAT



- Address Translation
 - All of the outgoing packet go through the NAT router which replaces the source address in the packet with the global NAT address
 - All incoming packet also pass through the NAT router which replaces the destination address in the packet with the appropriate private address

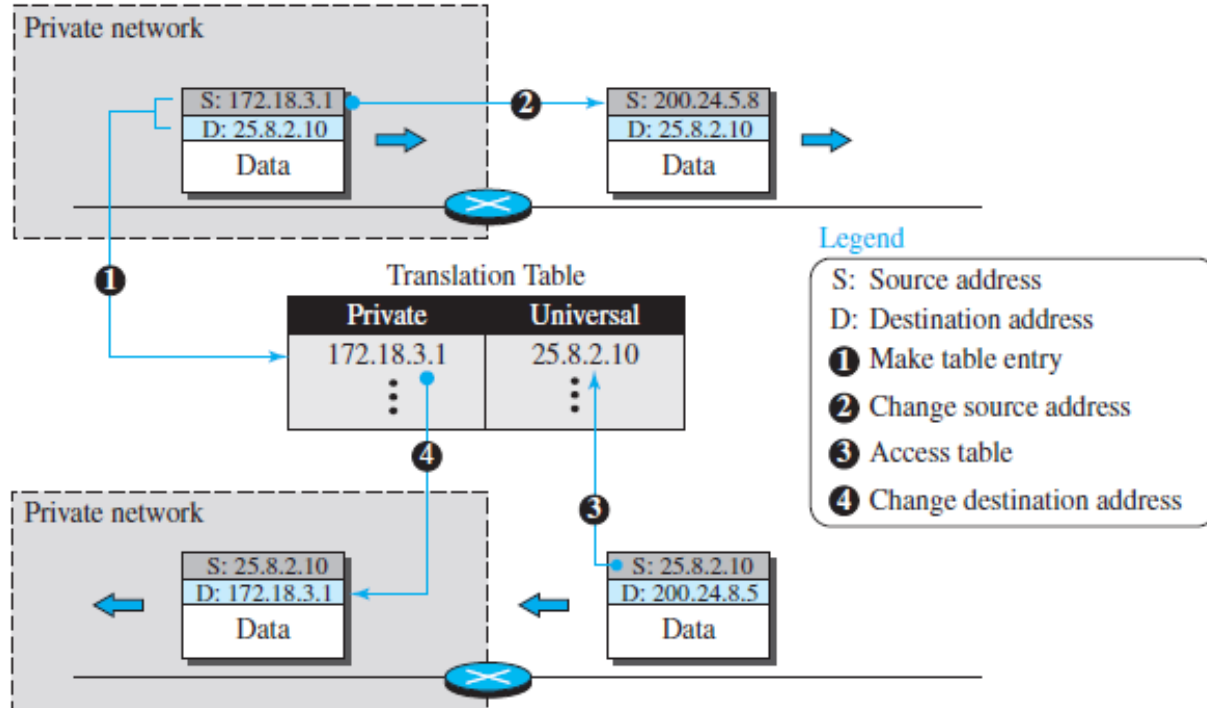


NAT



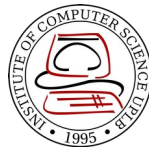
- Translation Table
 - Using one IP Address
 - Using a pool of IP Addresses
 - Using both IP Addresses and Port Numbers

NAT



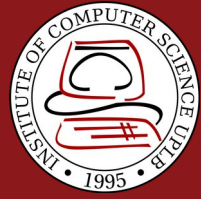
- Using a pool of IP address
 - Use of 1 global address by the NAT router allows only one private-network host to access a given external host
 - To remove restriction, NAT router can use a pool of global address
 - Example: 200.24.5.8, 200.24.5.9, 200.24.5.10, 200.24.5.11
 - 4 private-network host can communicate with the same external host at the same time because each pair of addresses defines a separate connection
 - However, no private-network host can access 2 external server programs at the same time

NAT



- Using both IP address and Port addresses

<i>Private address</i>	<i>Private port</i>	<i>External address</i>	<i>External port</i>	<i>Transport protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
⋮	⋮	⋮	⋮	⋮

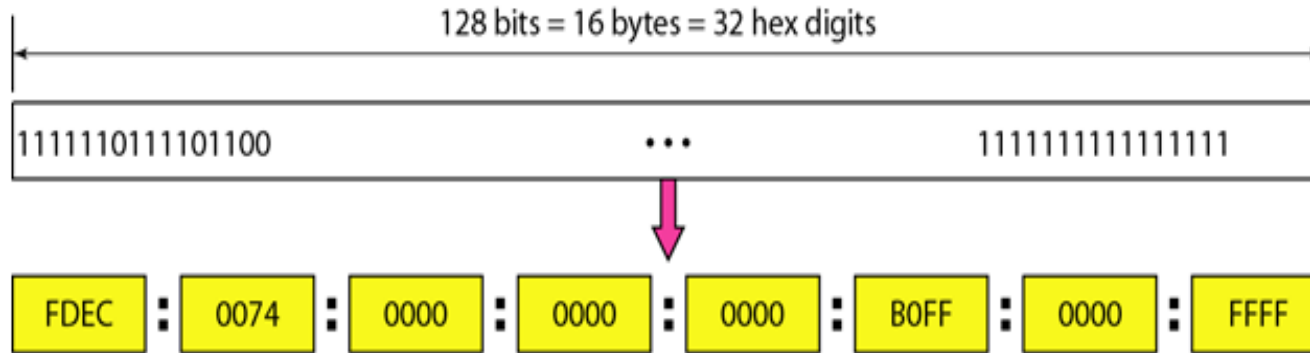


IPv6 Address

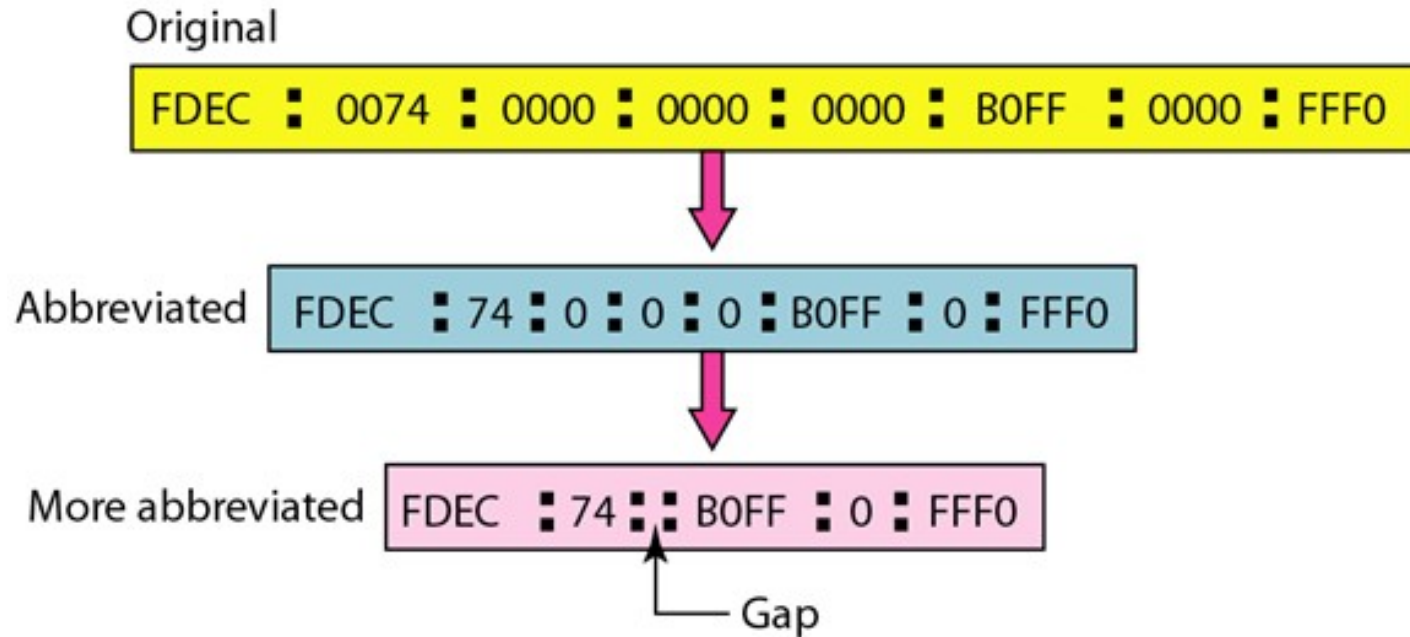
IPv6 Address



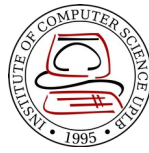
- RFC 2460
- 128-bit address (16 bytes/octets)
- Uses hexadecimal colon notation



IPv6 Address

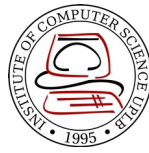


IPv6 Prefix



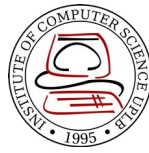
<i>Type Prefix</i>	<i>Type</i>	<i>Fraction</i>
0000 0000	Reserved	1/256
0000 0001	Unassigned	1/256
0000 001	ISO network addresses	1/128
0000 010	IPX (Novell) network addresses	1/128
0000 011	Unassigned	1/128
0000 1	Unassigned	1/32
0001	Reserved	1/16
001	Reserved	1/8
010	Provider-based unicast addresses	1/8

IPv6 Prefix

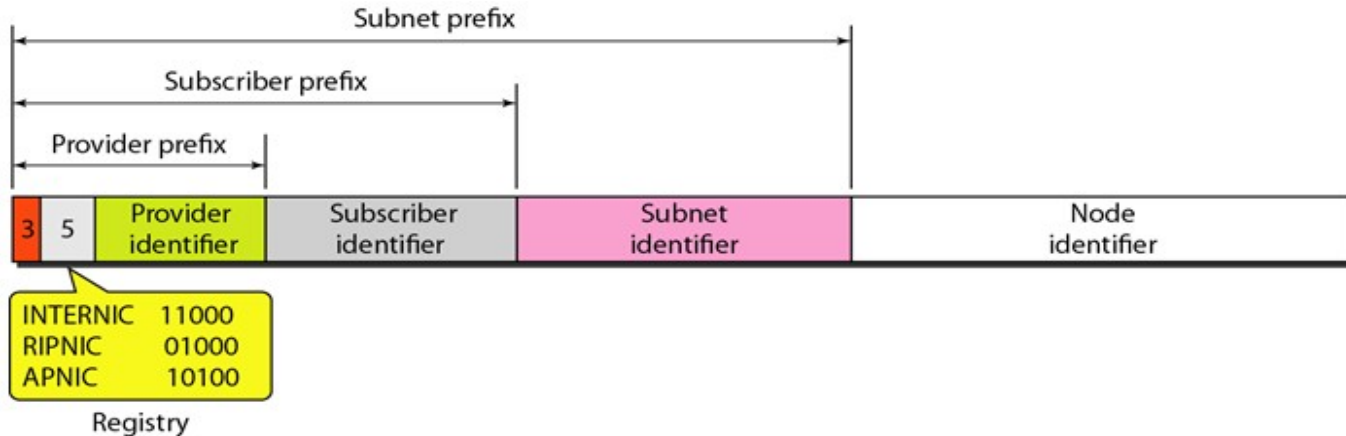


<i>Type Prefix</i>	<i>Type</i>	<i>Fraction</i>
011	Unassigned	1/8
100	Geographic-based unicast addresses	1/8
101	Unassigned	1/8
110	Unassigned	1/8
1110	Unassigned	1/16
1111 0	Unassigned	1/32
1111 10	Unassigned	1/64
1111 110	Unassigned	1/128
1111 1110 0	Unassigned	1/512
1111 1110 10	Link local addresses	1/1024
1111 1110 11	Site local addresses	1/1024
1111 1111	Multicast addresses	1/256

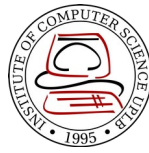
Unicast Address



- Defines a single computer
- Two types
 - Geographically based
 - Provider-based



Unicast Address

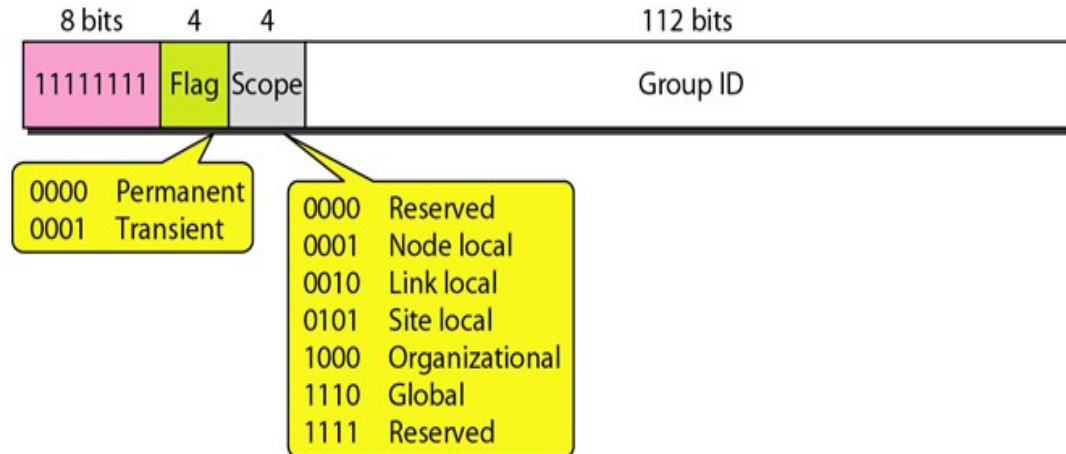


- **Type identifier** – 3-bit field that defines the address as a provider-based address
- **Registry identifier** – 5-bit field that indicates the agency that registered the address
- **Provider identifier** – variable-length field that identifies the provider for Internet access. (16-bit length recommended)
- **Subscriber identifier** – assigned to an organization when subscribed to the Internet through a provider (24-bit length recommended)
- **Subnet identifier** – defines a specific subnetwork under the territory of the subscriber (32-bit length recommended)
- **Node identifier** – identity of the node connected to a subnet (48-bit length recommended)

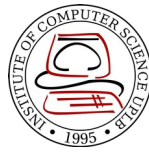
Multicast Address



- Defines a group of hosts
- A packet sent to a multicast address must be delivered to each member of the group

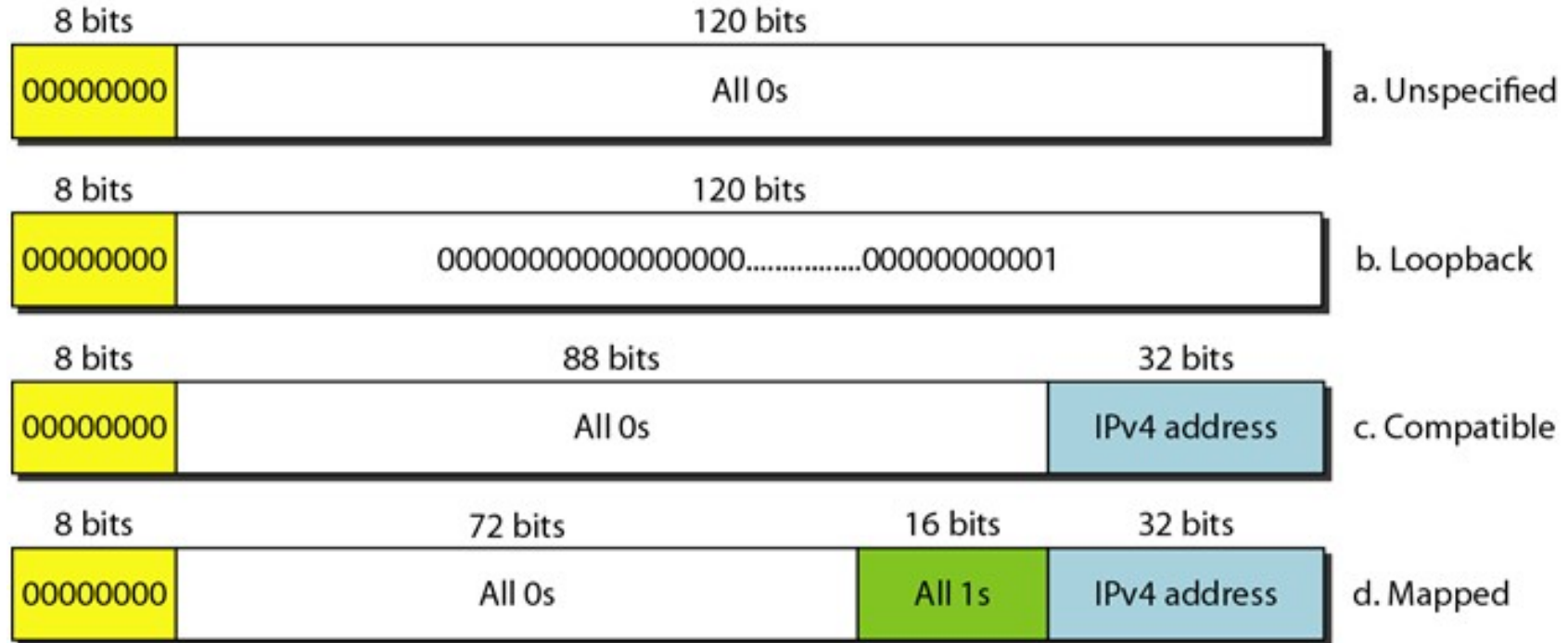
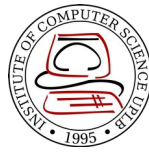


Anycast Address

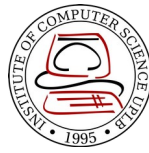


- Defines a group of nodes like multicast
- A packet destined for an anycast address is delivered to only one member of the anycast group, the nearest one.
- Possible use is to assign an anycast address to all routers of an ISP that covers a large logical area in the Internet

Reserved Address

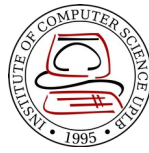


Reserved Address

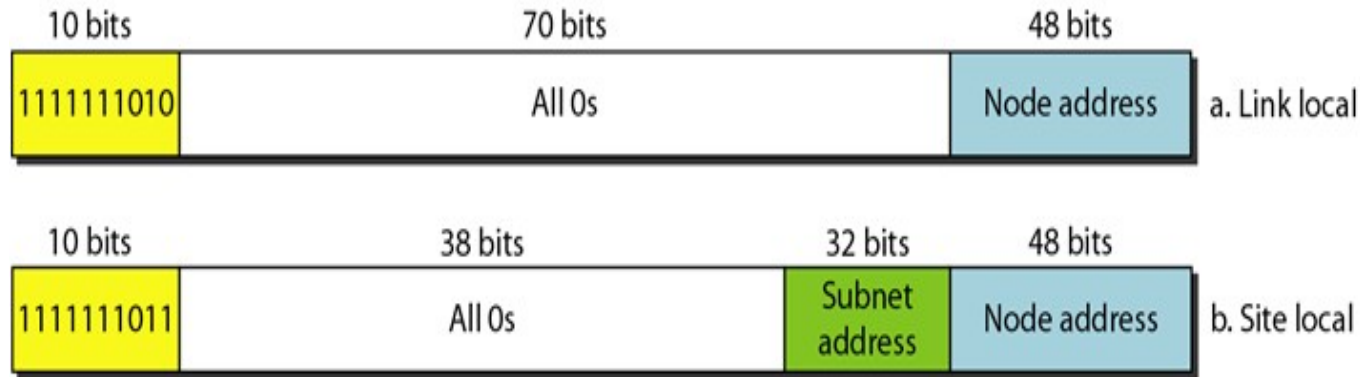


- **Unspecified address** – used when a host does not know its own address and sends an inquiry to find its address
- **Loopback address** – used by a host to test itself without going into the network
- **Compatible address** – used when a computer using IPv6 wants to send a message to another computer using IPv6, but the message needs to pass through a part of the network that still operates in IPv4
- **Mapped address** – used when a computer that has migrated to IPv6 wants to send a packet to a computer still using IPv4

Local Address



- Used when organization wants to use IPv6 protocol without being connected to the global Internet
 - Link local address – used in an isolated subnet
 - Site local address – used in an isolated site with several subnets



References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.